

GNIBEN, Denmark

WMO: 061690

Lat: 56.0083N

Lon: 11.2783E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.7	-3.5	-8.6	1.8	-3.4	-7.0	2.1	-2.5	18.1	3.1	16.3	2.6	6.2	140	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	23.5	19.7	22.2	18.9	21.0	18.1	20.5	22.6	19.4	21.6	18.5	20.6	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.3	21.9	18.5	13.4	20.9	17.6	12.6	20.0	58.9	22.6	55.4	21.6	52.5	20.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.8	13.5	DB	-5.5	26.2	2.0	1.5	-7.0	27.3	-8.2	28.1	-9.3	28.9	-10.7	30.0	(4)
(5)			WB	-6.5	21.3	1.9	1.2	-7.9	22.2	-9.1	22.8	-10.1	23.5	-11.5	24.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	1.6	1.3	3.2	7.0	11.7	15.0	17.6	17.8	14.8	10.6	6.5	3.5	(6)	
	DBStd	6.39	2.69	2.62	2.41	2.47	2.51	2.26	2.34	2.19	2.06	2.29	2.48	2.93	(7)	
	HDD10.0	1148	260	243	212	94	11	0	0	0	0	21	107	201	(8)	
	HDD18.3	3353	518	476	471	339	207	102	43	37	106	239	356	459	(9)	
	CDD10.0	883	0	0	0	5	62	151	236	242	146	39	2	0	(10)	
	CDD18.3	46	0	0	0	0	1	3	21	21	2	0	0	0	(11)	
	CDH23.3	49	0	0	0	0	1	2	25	22	1	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	7.0	8.1	7.3	6.7	6.0	6.0	6.5	6.0	6.6	7.2	7.6	8.0	8.3	(14)	
(15) Precipitation	PrecAvg	547	39	28	33	31	37	57	52	65	51	55	48	49	(15)	
	PrecMax	779	88	60	67	78	63	124	127	142	148	131	101	129	(16)	
	PrecMin	333	3	7	5	6	13	1	8	22	22	13	9	8	(17)	
	PrecStd	99	21	13	18	19	15	32	29	35	30	27	21	27	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	6.9	9.3	15.6	20.2	23.0	25.8	25.4	21.7	16.7	12.6	9.3	(19)	
		MCWB	7.0	5.9	7.3	12.3	16.4	19.3	20.1	20.5	17.7	14.3	11.6	8.4	(20)	
	2%	DB	6.9	6.0	7.7	13.2	18.0	20.9	23.6	23.7	19.9	15.1	11.3	8.5	(21)	
		MCWB	5.9	5.1	6.2	10.5	15.1	17.5	19.6	20.0	16.8	13.6	10.4	7.7	(22)	
	5%	DB	6.0	5.3	6.8	11.6	16.3	19.5	22.3	22.3	18.6	14.2	10.5	7.8	(23)	
		MCWB	5.3	4.7	5.6	9.5	13.9	16.5	19.0	19.2	16.2	12.8	9.6	7.1	(24)	
	10%	DB	5.2	4.7	6.0	10.3	15.1	18.3	21.1	21.1	17.5	13.5	9.7	7.2	(25)	
		MCWB	4.5	4.1	5.0	8.7	13.1	15.8	18.3	18.3	15.5	12.3	8.8	6.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	6.1	7.6	12.8	17.1	19.8	21.7	22.1	18.5	14.9	11.8	8.8	(27)	
		MCDB	7.7	6.5	8.8	15.0	19.5	22.2	23.9	24.0	20.5	15.8	12.3	9.1	(28)	
	2%	WB	6.2	5.4	6.6	10.8	15.5	18.1	20.4	20.8	17.4	13.8	10.6	7.9	(29)	
		MCDB	6.7	5.8	7.5	12.2	17.4	20.2	22.6	22.8	19.1	14.7	11.2	8.3	(30)	
	5%	WB	5.4	4.8	5.9	9.7	14.3	17.1	19.5	19.6	16.6	13.1	9.8	7.2	(31)	
		MCDB	5.9	5.2	6.6	11.0	15.8	18.9	21.6	21.7	18.2	13.9	10.4	7.7	(32)	
	10%	WB	4.6	4.2	5.2	8.7	13.4	16.1	18.7	18.7	15.7	12.5	8.9	6.6	(33)	
		MCDB	5.1	4.6	5.8	9.9	14.8	17.8	20.8	20.7	17.2	13.3	9.5	7.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	2.2	2.2	2.8	3.7	4.1	4.1	4.0	3.7	2.9	2.4	2.1	2.2	(35)	
		MCDBR	2.4	2.5	3.6	5.6	5.7	5.8	5.6	5.3	4.5	3.1	2.3	2.1	(36)	
	5% WB	MCWBR	2.5	2.2	2.7	3.7	3.8	4.1	3.5	3.3	3.0	2.5	2.4	2.3	(37)	
		MCWBR	2.3	2.3	3.5	4.8	5.1	5.4	5.0	4.7	4.1	3.0	2.2	2.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.315	0.338	0.359	0.394	0.388	0.380	0.402	0.402	0.374	0.355	0.335	0.303	(40)	
		taud	2.285	2.320	2.326	2.282	2.371	2.422	2.375	2.380	2.437	2.417	2.326	2.254	(41)	
	Ebn at Noon		583	721	800	823	853	864	834	808	781	696	553	496	(42)	
		Edh at Noon	54	76	95	115	111	107	110	103	86	69	52	44	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.43	1.02	2.46	4.11	5.34	5.77	5.34	4.24	2.98	1.47	0.56	0.31	(44)	
	RadStd		0.08	0.17	0.23	0.48	0.49	0.43	0.56	0.30	0.29	0.19	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-160	N/A	N/A

Nomenclature: See separate page